

HACKATHON 2

Implementation Results, Evaluation and Performance Analysis

TEAM 13

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1. Results Overview

The implemented system demonstrates an end-to-end customer-support RAG pipeline. The project successfully completed dataset preparation, embedding generation, FAISS index construction, retrieval, prompt construction, Gemini integration testing, FastAPI backend implementation, React frontend implementation, and frontend-to-backend communication.

2. Implementation Results

Area	Result / Status
Dataset Collection	Completed
Dataset Processing	Completed
Sentence Embeddings	Implemented
all-MiniLM-L6-v2	Implemented
FAISS Vector Index	Implemented
Top-K Retrieval	Implemented
RAG Pipeline	Implemented
Prompt Construction	Implemented
Gemini Integration	Implemented and tested
FastAPI Backend	Running
React + Vite Frontend	Running
Frontend ↔ Backend Communication	Successful
POST /chat	Successful during application testing
Source-Aware Response	Implemented

3. Retrieval Evaluation

The project evaluation used the 200-record FAQ dataset. Retrieval-related metrics were calculated independently of the final Gemini generation stage.

Metric	Result	Interpretation
Precision@5	68.40%	Proportion of the top-5 retrieved records that were relevant.
Recall@5	100.00%	Relevant records were retrieved within the top-5 under the evaluation setup.
MRR	99.29%	The first relevant result appeared at a high rank under the evaluation setup.
Context Relevance	100.00%	Retrieved context was judged relevant under the project evaluation.

4. Component-Level Performance

Component	Score	Status / Basis
all-MiniLM-L6-v2	99.72%	Project retrieval/embedding evaluation result
FAISS Retrieval	~90.19%	Project-specific composite retrieval score
RAG Context	100.00%	Context relevance evaluation
Gemini Generation	~92.00%*	Provisional estimate
Overall PA-RAG	~93.17%*	Provisional project-specific composite

* The Gemini Generation and Overall PA-RAG values are provisional estimates, not final experimentally measured end-to-end accuracy. The full Gemini evaluation was blocked by the available API quota. Therefore, these values should not be presented as verified final accuracy in an academic results table.

5. Overall Score Calculation

The provisional overall score was calculated using a project-specific weighted composite. It is not a standardized RAG benchmark.

Overall = $0.20 \times \text{Embedding Score} + 0.25 \times \text{FAISS Score} + 0.15 \times \text{RAG Context Score} + 0.40 \times \text{Gemini Score}$

Using the provisional component values, the resulting project-specific composite is approximately 93.17%.

8. Retrieval Quality Interpretation

The retrieval results indicate that the evaluation setup was able to retrieve relevant support information effectively. Recall@5 of 100% means that the relevant target information was found within the top five retrieved records under the project's evaluation methodology. MRR of 99.29% indicates that the first relevant result generally appeared at a very high rank. Precision@5 of 68.40% indicates that not every item in the top five was relevant.

9. Literature Comparison

The literature comparison contains reported or derived metric values from selected research papers. Because the papers use different datasets, tasks, metrics and evaluation protocols, their values are not directly equivalent universal accuracy scores.

Research Approach	Comparison Value	Metric Basis
RAG – Lewis et al.	49.68%	Mean RAG-Sequence EM across selected benchmarks
DPR – Karpukhin et al.	74.80%	Mean Top-20 retrieval accuracy
REALM – Guu et al.	42.63%	Mean selected open-domain QA EM
RAG Best Practices – Wang et al.	48.30%	Reported overall average
RAGAs – Es et al.	81.00%	Mean agreement accuracy for selected metrics
SELF-ROUTE – Li et al.	46.41	Average benchmark score
Enhanced RAG – González Juclà et al.	64.20%	Mean Recall@10 across selected datasets
KG-RAG – Xu et al.	91.86%	Mean of selected retrieval metrics

Proposed PA-RAG	~93.17%*	Provisional project-specific composite
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Several papers in the literature survey do not report a single comparable accuracy value, so they should not be assigned zero merely to create a graph.

11. Results That Must Be Clearly Labeled

- Gemini Answer Relevance ~92% — provisional estimate.
- Gemini Faithfulness ~92% — provisional estimate.
- Overall PA-RAG ~93.17% — provisional composite, not a standardized accuracy.
- Do not report the failed-run 57.52% as final accuracy.

14. Conclusion

The implemented project successfully demonstrates the core components of an LLM-based customer-support RAG system. The strongest currently verified results are the retrieval metrics: Precision@5 of 68.40%, Recall@5 of 100.00%, MRR of 99.29%, and Context Relevance of 100.00%. The application stack is also operational across the FAISS retrieval layer, FastAPI backend, and React frontend. Final generation-quality metrics remain to be measured after sufficient Gemini API evaluation capacity is available.